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CS-499

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5-1 Journal: Significant Trends in CS

Part 1:

1. What is the significance of each trend?
 - a. AI and Quantum Computing are two major trends that are significant if not a major paradigm shift in the tech industry that are building momentum currently.

AI will likely be very transformative in the industry in the next few years.

Quantum Computing is making major advancement with Google and IBM, with fears of “Q-day” starting to build.
2. How will each trend change the field of computer science?
 - a. AI likely will act as a serious augmentation for developers as the models improve their ability to edit and write code as well as automate tasks. Quantum Computing could cause all of our encryption algorithms to simultaneously become useless and cause the opportunity for apocalyptic data theft. The latter requires extensive research on the development of Quantum Proof encryption that will be ready prior to the emergence of commercial Quantum Computers.
3. How will each trend change the experience of consumers, workers, or citizens?
 - a. Consumers will likely not see much from quantum computing at first, but large corporations and nation states could gain massive capabilities. For AI all people

have already been impacted in some way. Mainly, we have the ability to get answers to many hard questions and create content and papers.

4. How will each trend fit in with your career interests or aspirations?
 - a. Currently, IBM (Red Hat's parent company) has some Quantum projects and we at Red Hat are gearing up for AI deployment in our products. So the impact will be profound likely in the near future.
5. Which course outcomes have you achieved so far, and which ones remain?
 - a. Outcomes 2, 3, 4 and 5

Part 2:

Status Checkpoints for All Categories

Checkpoint	Software Design and Engineering	Algorithms and Data Structures	Databases
Name of Artifact Used	Appointment (CS-320)	Course Catalog (CS-300)	Animal Search DB (CS-340)
Status of Initial Enhancement	Complete	Complete	Complete
Submission Status	Submitted	Submitted	Submitted
Status of Final Enhancement	Complete	Complete	Complete

Uploaded to ePortfolio	Complete	Complete	Pending
Status of Finalized ePortfolio	Pending	Pending	Pending